Beam Mismatch and Halo Formation

USPAS - Beam Physics with Intense Space Charge

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2D envelope modes in continuous focusing channels

Derivation of equilibrium beam radius

KV envelope equations describe evolution of elliptical beam radii $r_{x,y}$ in linear focusing system:

$$
\begin{aligned}
r_x'' + \kappa_x(s) r_x - \frac{2Q}{r_x + r_y} - \frac{\varepsilon_x^2}{r_x^3} &= 0, \\
r_y'' + \kappa_y(s) r_y - \frac{2Q}{r_x + r_y} - \frac{\varepsilon_y^2}{r_y^3} &= 0,
\end{aligned}
\tag{1}
$$

where $\varepsilon_{x,y}$ are invariant emittances, Q is beam perveance, and $\kappa_{x,y}(s)$ are linear focusing strengths.

Restrict attention to continuous, axisymmetric focusing systems with $\kappa_x(s) = \kappa_y(s) = k_0^2$. Also assume equal emittances: $\varepsilon_x = \varepsilon_y = 0$.

$$
\begin{aligned}
r_x'' + k_0^2 r_x - \frac{2Q}{r_x + r_y} - \frac{\varepsilon^2}{r_x^3} &= 0, \\
r_y'' + k_0^2 r_y - \frac{2Q}{r_x + r_y} - \frac{\varepsilon^2}{r_y^3} &= 0.
\end{aligned}
\tag{2}
$$

Equilibrium occurs when $r_x'' = r_y'' = 0$. The equilibrium radius $r_x = r_y = r_0$ solves:

$$k_0^2 r_0 - Q r_0^{-1} - \varepsilon_2 r_0^{-3} = 0. \tag{3}$$

$$r_0 = \frac{Q^{1/2}}{k_0} \left[\frac{1}{2} + \frac{1}{2} \sqrt{1 + \frac{4k_0^2 \varepsilon^2}{Q^2}} \right]^{1/2}. \tag{4}$$

In limit of space-charge-dominated ($2\kappa_0 \varepsilon \ll Q$), or emittance-dominated ($2\kappa_0 \varepsilon \gg Q$) beam,

$$r_0 \to \begin{cases} \sqrt{Q}/k_0, & 2k_0\varepsilon \ll Q, \\ \sqrt{\varepsilon/k_0}, & 2k_0\varepsilon \gg Q. \end{cases} \tag{5}$$

Definition of depressed betatron frequency

Single-particle equations of motion within beam edge:

$$\begin{aligned} x'' + k_0^2 x - \frac{2Q}{r_x + r_y}\frac{x}{r_x} &= 0, \\ y'' + k_0^2 y - \frac{2Q}{r_x + r_y}\frac{y}{r_y} &= 0. \end{aligned} \tag{6}$$

At equilibrium ($r_x = r_y = r_0$),

$$\begin{aligned} x'' + \left(k_0^2 - \frac{Q}{r_0^2}\right) x &= 0, \\ y'' + \left(k_0^2 - \frac{Q}{r_0^2}\right) y &= 0. \end{aligned} \tag{7}$$

Define *depressed* focusing strength $k^2 \equiv k_0^2 - Q/r_0^2$:

$$
\begin{aligned}
x'' + k^2 x &= 0, \\
y'' + k^2 y &= 0.
\end{aligned}
\tag{8}
$$

Perturbation of equilibrium distribution

Define deviations from the equilibrium radii:

$$\begin{aligned} r_x(s) &= r_0 + \eta_x(s), \\ r_y(s) &= r_0 + \eta_y(s). \end{aligned} \tag{9}$$

Substituting (9) into (**??**) gives:

$$\begin{aligned} \eta_x'' + k_0^2(r_0 + \eta_x) - \frac{Q}{r_0}\left(1 + \frac{\eta_x + \eta_y}{2r_0}\right)^{-1} - \frac{\varepsilon^2}{r_0^3}\left(1 + \frac{\eta_x}{r_0}\right)^{-3} &= 0, \\ \eta_y'' + k_0^2(r_0 + \eta_y) - \frac{Q}{r_0}\left(1 + \frac{\eta_x + \eta_y}{2r_0}\right)^{-1} - \frac{\varepsilon^2}{r_0^3}\left(1 + \frac{\eta_y}{r_0}\right)^{-3} &= 0. \end{aligned} \tag{10}$$

Assume deviations $\eta_{x,y}$ are much smaller than equilibrium radius r_0. Then linearize perturbed envelope equations (using $(1+x)^r \approx 1 + rx$. This gives:

$$\begin{aligned}
\eta_x'' + k_0^2(r_0+\eta_x) - \frac{Q}{r_0}\left(1 - \frac{\eta_x+\eta_y}{2r_0}\right) - \frac{\varepsilon^2}{r_0^3}\left(1 - 3\frac{\eta_x}{r_0}\right) = 0, \\
\eta_y'' + k_0^2(r_0+\eta_y) - \frac{Q}{r_0}\left(1 - \frac{\eta_x+\eta_y}{2r_0}\right) - \frac{\varepsilon^2}{r_0^3}\left(1 - 3\frac{\eta_y}{r_0}\right) = 0.
\end{aligned} \tag{11}$$

Factoring η_x and η_y gives:

$$\begin{aligned}
\eta_x'' + \left(k_0^2 + \frac{Q}{2r_0^2} + \frac{3\varepsilon^2}{r_0^4}\right)\eta_x + \left(\frac{Q}{2r_0^2}\right)\eta_y = 0, \\
\eta_y'' + \left(k_0^2 + \frac{Q}{2r_0^2} + \frac{3\varepsilon^2}{r_0^4}\right)\eta_y + \left(\frac{Q}{2r_0^2}\right)\eta_x = 0,
\end{aligned} \tag{12}$$

where we have used **?@eq-equilibrium-radius-equation** to eliminate the remaining terms.

Using the definition of k (**?@eq-depressed-focusing-strength**) we find:

$$
\begin{aligned}
\eta_x'' + \left(\frac{3}{2}k_0^2 + \frac{5}{2}k^2\right)\eta_x + \left(\frac{1}{2}k_0^2 - \frac{1}{2}k^2\right)\eta_y = 0, \\
\eta_y'' + \left(\frac{3}{2}k_0^2 + \frac{5}{2}k^2\right)\eta_y + \left(\frac{1}{2}k_0^2 - \frac{1}{2}k^2\right)\eta_x = 0.
\end{aligned}
\tag{13}
$$

Eigenmode analysis or perturbed envelope equations

(13) defines a coupled linear system of equations for $\eta_{x,y}$. We will analyze this system in terms of its normal modes. In vector form, **?@eq-env-pert-coupeld-1** can be written:

$$\eta'' + \mathbf{A}\eta = 0, \tag{14}$$

where $\eta = [\eta_x, \eta_y]^T$ is the vector of perturbed envelope radii and $\mathbf{A}$ is a matrix of coefficients:

$$\mathbf{A} = \begin{bmatrix} \alpha_1 & \alpha_2 \\ \alpha_2 & \alpha_1 \end{bmatrix}, \tag{15}$$

with entries defined as:

$$\begin{aligned} \alpha_1 &= \frac{3}{2}k_0^2 + \frac{5}{2}k^2, \\ \alpha_2 &= \frac{1}{2}k_0^2 - \frac{1}{2}k^2. \end{aligned} \tag{16}$$

We search for solutions of the form

$$\eta_{\pm}(s) = \eta_0 e^{ik_{\pm}s}, \tag{17}$$

where $k_{\pm}$ are two distinct frequencies. This leads to an eivenvalue problem:

$$\left(\mathbf{A} - \mathbf{I}k_{\pm}^2\right)\eta_{\pm} = 0. \tag{18}$$

The eigenvalues correspond to the following frequencies, written in terms of the natural frequency k_0 and depressed frequency k, are:

$$\begin{aligned} k_+ &= \alpha_1 + \alpha_2 = 2k_0^2 + 2k^2. \\ k_- &= \alpha_1 - \alpha_2 = k_0^2 + 3k^2. \end{aligned} \tag{19}$$

The frequencies correspond to eigenvectors $\eta_\pm$:

$$\begin{aligned} \eta_+ &= \sqrt{\frac{1}{2}} \begin{bmatrix} +1 \\ +1 \end{bmatrix} \quad \text{(breathing mode)}. \\ \eta_- &= \sqrt{\frac{1}{2}} \begin{bmatrix} +1 \\ -1 \end{bmatrix} \quad \text{(quadrupole mode)}. \end{aligned} \tag{20}$$

Mode frequency vs. tune depression

[Plots showing mode frequency vs. tune depression.]

Breathing vs. quadrupole mode

[Plots showing breathing vs. quadrupole mode.]

3D envelope modes in continuous focusing channels

It is natural to extend the above analysis to three-dimensional systems. We will simplify things by restricting our attention to upright beams with axial symmetry. We study the coupling between the transverse radius $r_\perp$ and the longitudinal radius r_z.

[...]

Particle-core models of halo formation

Measured phase space distribution in SNS linac

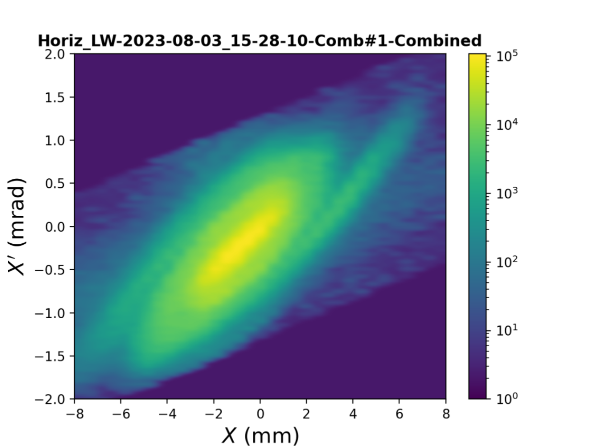

Figure 1

What is halo?

- Vague definition: low-density cloud surrounding dense core.
- Important for high-power accelerators (beam loss).
- Difficult to measure and predict.

Particle-core models of halo formation

- Describe motion of “test” particles in field of lattice + oscillating beam core.
- Much simpler than self-consistent Vlasov-Poisson (V-P) model, but useful for isolating halo formation mechanisms.
- Basic features of particle-core models show up in PIC simulations.

Plan of attack

- Start with KV equilibria in two-dimensional, axisymmetric, continuous focusing systems.
- Particles within beam experience *linear* forces and remain there for all time. (Even with mismatch!)
- Introduce mismatch of envelope, such that particles can move in/out of the core.

Return to **?@fig-mode-freq** which plots the envelope mode frequencies $k_{\pm}$ as a function of the depressed single-particle frequency (a measure of space charge intensity). Particles far from the core oscillate at the natural frequency k_0, which is determined by the external focusing strength. Particles inside the core oscillate at the depressed frequency $k \leq k_0$. Outside the core, the oscillation frequency decreases continuously and nonlinearly with the radius r. We expect resonances to occur when the particle oscillation frequency is a multiple of the envelope frequency $k_{\pm}$—particularly when the envelope frequency is *twice* the single-particle frequency. We also expect resonances to be enhanced by larger oscillations of the beam. [...]

This section will focus on an important paper written by Gluckstern in 1994 [1]. At the time, the maximum halo extent was an important theoretical question—and an important practical question for the design of new high-power accelerator. Gluckstern used classical perturbation theory to derive an approximate Hamiltonian to describe the resonant particle dynamics and, consequently, an upper bound on the halo amplitude. The Hamiltonian contours form the famous "peanut diagram'' and is reproduced quite well by PIC simulations. [...]

Equations of motion

Consider a matched KV distribution of radius R in a continuous focusing channel. The single-particle equation of motion in the horizontal plane is

$$x'' + k_0^2 x = \begin{cases} \frac{Q}{R^2}x, & r \leq R, \\ \frac{Q}{r^2}x, & r \geq R, \end{cases} \tag{21}$$

where $r^2 = x^2 + y^2$. (The equation is the same in the vertical plane.) For convenience, we will rewrite the equations using the step function function Θ:

$$x'' + k_0^2 x = \underbrace{\left(\frac{Q}{r^2}\right) x\Theta(r - R)}_{\text{Exterior}} + \underbrace{\left(\frac{Q}{R^2}\right) x\Theta(R - r)}_{\text{Interior}}, \tag{22}$$

where

$$\Theta(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0. \end{cases} \tag{23}$$

Define a sinusoidal perturbation of the beam envelope radius:

$$R(s) + R_0 \left(1 - \epsilon \cos(k_+ s)\right), \tag{24}$$

where k_+ is the breathing mode frequency, R_0 is the equilibrium radius, and ϵ is the strength of the perturbation. Subtracting the linear defocusing force Qx/R_0^2 from both sides gives

$$x'' + k^2 x = -\frac{Q}{R_0^2}\left(1 - \frac{R_0^2}{r^2}\right) x\Theta(r - R) + Q\left(\frac{1}{R^2} - \frac{1}{R_0^2}\right) x\Theta(R - r), \tag{25}$$

where k^2 is the depressed oscillation frequency within the beam:

$$k^2 = k_0^2 - \frac{Q}{R_0^2}. \tag{26}$$

If ϵ is small, the $1/R^2$ term can be approximated by truncating its Taylor series expansion:

$$R^{-2} = R_0^{-2}\left(1 - \epsilon\cos(k_+ s)\right)^{-2} \approx R_0^{-2}\left(1 + 2\epsilon\cos(k_+ s)\right). \tag{27}$$

Substituting the approximation into the equation of motion gives:

$$x'' + k^2 x = -\frac{Q}{R_0^2}\left(1 - \frac{R_0^2}{r^2}\right) x\Theta(r - R) + \frac{2\epsilon Q}{R_0^2} x\cos(k_+ s)\Theta(R - r). \tag{28}$$

The first term is a nonlinear force that gives rise to an amplitude-dependent frequency. The second term is a linear force arising from the envelope perturbation. This term allows the transfer of particles from the interior to the exterior of the hard-edged beam core.

We will make two further approximations in our analysis. First, we will assume there is no angular momentum and simply the one-dimensional oscillations along the x axis. Second, we will approximate the nonlinear defocusing outside the beam using a *pendulum model*:

$$x'' + k^2 x = -\frac{Q}{R_0^4} x^3 + \frac{2\epsilon Q}{R_0^2} x \cos(k_+ s). \tag{29}$$

Note that the second term (core oscillation) now extends beyond the core. While the pendulum model is a crude approximation of the external nonlinearity, it can still capture the essential physics. We will return to the full model later on.

Phase-amplitude coordinates

To analyze the equations of motion in **?@eq-pcm-eom-pendulum**}, we transform to amplitude-phase coordinates A, ψ:

$$\frac{x(s)}{R_0} \equiv A(s)\sin(\psi(s)), \tag{30}$$

$$\frac{x'(s)}{R_0} \equiv kA(s)\cos(\psi(s)), \tag{31}$$

with the phase $\psi(s)$ defined as

$$\psi(s) = ks + \alpha(s), \tag{32}$$

where ks is the unperturbed phase advance within the beam and $\alpha(s)$ is a slowly varying function. We will derive dynamical equations for A and ψ by substituting **?@eq-pcm-phase-amp-x** and **?@eq-pcm-phase-amp-xp** into **?@eq-pcm-eom-pendulum**.

First, note that the s-derivative of **?@eq-pcm-phase-amp-x** gives

$$\frac{x'}{R_0} = A' \sin\psi + A(k + \alpha') \cos\psi. \tag{33}$$

Therefore, from the definition in **?@eq-pcm-phase-amp-x**,

$$A' \sin\psi + \alpha' A \cos\psi = 0. \tag{34}$$

The derivative of x' in **?@eq-pcm-phase-amp-xp** gives

$$\frac{x''}{R_0} = kA' \cos\psi - kA(k + \alpha') \sin\psi. \tag{35}$$

The equation of motion **?@eq-pcm-eom-pendulum** can then be written as

$$R_0 k \left(A' \cos\psi - A\alpha' \sin\psi\right) = f(A, \psi, s), \tag{36}$$

where $f(A, \psi, s)$ is the driving force on the right hand side of **?@eq-pcm-eom-pendulum** expressed in the new coordinates:

$$f(A, \psi, s) = \frac{Q}{R_0} \left[-A^3 \sin^3\psi + 2\epsilon A \sin\psi \cos(k_+ s)\right]. \tag{37}$$

?@eq-pcm-deriv-x and **?@eq-pcm-eom-pendulum-amp-phase**} define a linear system to solve for A' and α'.

$$\begin{bmatrix} kR_0 \cos\psi & -kR_0 A \sin\psi \\ \sin\psi & A\cos\psi \end{bmatrix} \begin{bmatrix} A' \\ \alpha' \end{bmatrix} = \begin{bmatrix} f \\ 0 \end{bmatrix}. \tag{38}$$

Solving the linear system gives the dynamics of the amplitude and phase:

$$.A' = \frac{f(A, \psi, s)}{kR_0} \cos\psi, \tag{39}$$

$$-\alpha' = \frac{f(A, \psi, s)}{kR_0 A} \sin\psi. \tag{40}$$

Phase averaging

We expect resonance to occur when the envelope oscillation frequency k_+ is a multiple of the particle oscillation frequency k. Therefore, we define a shifted phase coordinate Ψ which will vary slowly near the resonant frequency:

$$\Psi \equiv 2\psi - k_+ s. \tag{41}$$

Using this new variable, the driving force in **?@eq-pcm-amp-phase-drive** becomes

$$f(A, \Psi, \psi) = \frac{Q}{R_0}\left[-A^3 \sin^3\psi + 2\epsilon A \sin\psi \cos(2\psi - \Psi)\right], \tag{42}$$

and the full expressions for A' and α' become:

$$\begin{aligned} A' &= \frac{Q}{kR_0^2}\left[A^3 \sin^3\psi \cos\psi + 2\epsilon A \sin\psi \cos\psi \cos(2\psi - \Psi)\right] \\ \alpha' &= \frac{Q}{kR_0^2}\left[A^2 \sin^4\psi - 2\epsilon \sin^2\psi \cos(2\psi - \Psi)\right] \end{aligned} \tag{43}$$

We now average the expressions **?@eq-pcm-dyn** over one period:

$$
\begin{aligned}
A' &\to \frac{1}{2\pi}\int_0^{2\pi} A'\,d\psi = \frac{\epsilon Q}{2kR_0^2} A\sin\Psi, \\
\alpha' &\to \frac{1}{2\pi}\int_0^{2\pi} \alpha'\,d\psi = \frac{3Q}{8kR_0^2} A^2 + \frac{\epsilon Q}{2kR_0^2}\cos\Psi.
\end{aligned} \tag{44}
$$

The resonant phase Ψ is held constant during the integration.

Switching to variables $w = A^2/R_0^2$ and $\Psi' = (2k - k_+) + 2\alpha'$:

$$
\begin{aligned}
w' &= \frac{\epsilon Q}{kR_0^2} w \sin\Psi, \\
\Psi' &= \frac{Q}{kR_0^2}\left[-\Delta + \frac{3w}{4} + \epsilon\cos\Psi\right],
\end{aligned}
\tag{45}
$$

where

$$
\Delta = k(k_+ - 2k)\frac{R_0^2}{Q}. \tag{46}
$$

The variables w and Ψ are canonically conjugate: their evolution equations can be derived from a Hamiltonian H:

$$
\begin{aligned}
H(w,\Psi) &= \frac{Q}{kR_0^2}\left[\Delta w - \frac{3}{8}w^2 - \epsilon w\cos\Psi\right], \\
w' &= \partial H/\partial\Psi, \\
\Psi' &= \partial H \partial w.
\end{aligned}
\tag{47}
$$

Analysis of the Hamiltonian

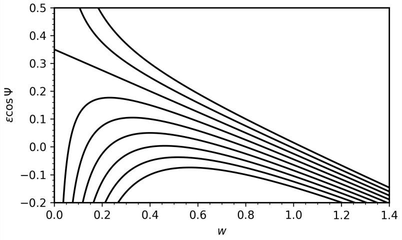

Figure 2: To do: limit range from -ϵ to ϵ.

Hamiltonian contours (polar coordinates)

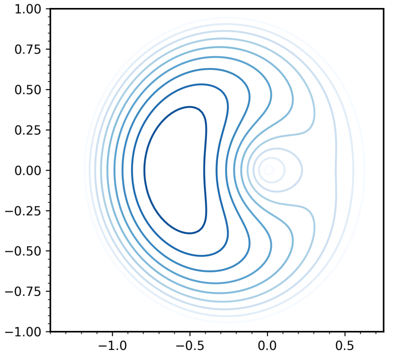

Figure 3: Hamiltonian contours as function of $w \cos \Psi$ and $w \sin \Psi$.

Hamiltonian contours (phase space coordinates)

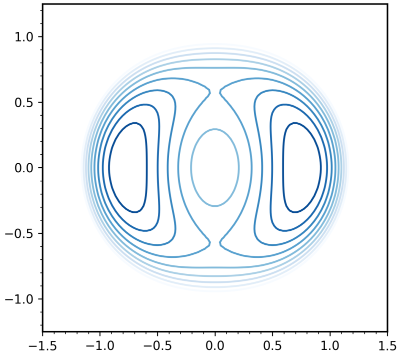

Figure 4: Hamiltonian contours as function of $x \propto \sqrt{w}\sin\Psi$ and $x' \propto \sqrt{w}\cos\Psi$.

Comparison to exact model

The exact (hard-edged) model in **?@eq-pcm-eom-exact** can be treated in the same way. [...]

Particle tracking results

[This will follow the beginning of Barnard lecture (with all the pictures).]

Three-dimensional particle-core models

[Will cover longitudinal stuff in Barnard lecture.]

Summary

[...]

References

[1] R. L. Gluckstern, *Analytic Model for Halo Formation in High Current Ion Linacs*, Phys. Rev. Lett. **73**, 1247 (1994).